Question Paper

Exam Date & Time: 07-Jan-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech III Semester MAKEUP Examination DEC/JAN 2024/25
COMPUTER SCIENCE & FINANCIAL TECHNOLOGY

MATHEMATICS FOR FINANCE [MAT 2132]

Marks: 50 **Duration: 180 mins.**

Answer all the questions. Section Duration: 180 mins

Answer all questions

1A) Find a spanning set for the null space of the matrix $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$ (3)

1B) Determine whether $v = \begin{bmatrix} -2 \\ 5 \\ 1 \end{bmatrix}$ is in the column space of $A = \begin{bmatrix} 2 & 5 & 1 \\ -1 & -7 & -5 \\ 3 & 4 & -2 \end{bmatrix}$ (3)

1C) Find the LU factorization of the matrix $A = \begin{bmatrix} +2 & -4 & -2 & +3 \\ +6 & -9 & -5 & +8 \\ +2 & -7 & -3 & +9 \\ +4 & -2 & -2 & -1 \\ -6 & +3 & +3 & +4 \end{bmatrix}$ (4)

2A) Apply the Gram-Schmidt algorithm to orthogonalize the basis{(1,1,1),(1,0,1),(1,1,2)} (3)

2B) Determine the definiteness of the quadratic form $5x^2 - y^2 + 7z^2 + 5xy - 3xz$ (3)

2C) Find the singular value decomposition of the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ (4)

3A) Determine if $A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$ is diagonalizable (3)

3B) Solve the difference equation $a_{n+2} - 7a_{n+1} + 12a_n = 0, a_0 = 0, a_1 = -1$. Use the method of generating function. (3)

3C) Find the least squares solution of $Ax = b$, where $A = \begin{bmatrix} 1 & -3 & -3 \\ 1 & 5 & 1 \\ 1 & 7 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 5 \\ -3 \\ -5 \end{bmatrix}$ and compute the associated least squares error. (4)

4A) (3)

Solve $y'' + y + 1 = 0$, $y(0) = 0$, $y(1) = 0$, with $h = 0.5$

4B) (3)

Find a Fourier series expansion for $f(x) = \begin{cases} 0, -\pi < x < 0 \\ \pi, 0 < x < \pi \end{cases}$

4C) (4)

Solve $u_{xx} + u_{yy} = -81xy$, in $0 < x < 1$, $0 < y < 1$, given the boundary conditions $u(x,0) = u(0,y) = 0$, $u(x,1) = 100$, $u(1,y) = 100$. Take $h = k = 1/3$. Use Gauss-Seidel method for solving. Perform two iterations

5A) (3)

$$f(x) = \begin{cases} 1, |x| \leq 1 \\ 0, |x| > 1 \end{cases}$$

Find the Fourier transform of

5B) (3)

Obtain the first four terms in a series of sines to represent T

θ°	0	30	60	90	120	150
T	0	5224	8097	7850	5499	2626

5C) (4)

Find the solution of the equation $\frac{\partial u}{\partial t} = \frac{1}{16}\frac{\partial^2 u}{\partial x^2}$, $0 < x < 1$, $t > 0$, subject to the conditions $u(x,0) = 0$, $u(0,t) = 0$ and $u(1,t) = 100t$. Use Crank-Nicolson scheme. Take $\lambda = 1$ and $h = 1/4$. Compute for one step.

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